10th grade
Learning evidence 2.1: Analysis of Decisions

Name: ____

Evaluated criteria

This exam evaluates only the criteria C1–C5. Each criterion is evaluated across the three problems below. A criterion is passed if it is correctly activated in at least two of the three problems. The overall exam result is binary (Passed/Not passed).

- C2 Interprets decision alternatives, events, consequences, and states of nature.
- C3 Builds a payoff table from a description of the problem.
- C4 Explains the maximax criterion for decision making without probabilities.
- C5 Summarizes the maximin criterion for decision making without probabilities.
- C1 Describes the way a problem can be formulated for optimal decision making.

Problem 1: A school cafeteria must choose one of three menu plans for a two-week festival: Plan A (local produce focus), Plan B (mixed menu), or Plan C (bulk staples). Demand for meals can be high, medium, or low. Based on past festivals, the probability of high demand is $0{,}35$, medium demand is $0{,}45$, and low demand is $0{,}20$. Profits are measured in hundreds of dollars. If demand is high, Plan A yields 92, Plan B yields 70, and Plan C yields 55. If demand is medium, Plan A yields 50, Plan B yields 62, and Plan C yields 48. If demand is low, Plan A yields -8, Plan B yields 30, and Plan C yields 38. The stated probabilities must be used for an expected value comparison as part of the decision analysis.

Problem 2: A delivery cooperative is choosing a fleet strategy for the next season: Strategy A (electric vans), Strategy B (hybrid vans), or Strategy C (diesel vans). Demand for deliveries can be strong, steady, or weak, and fuel costs can be low or high. The cooperative estimates the demand probabilities as $0{,}30$ (strong), $0{,}50$ (steady), and $0{,}20$ (weak). Fuel costs are independent of demand, with probabilities $0{,}55$ (low) and $0{,}45$ (high). Revenue, in thousands of dollars, depends on the demand level and strategy. Under strong demand, revenue is 420 for Strategy A, 380 for Strategy B, and 340 for Strategy C. Under steady demand, revenue is 320 for Strategy A, 300 for Strategy B, and 270 for Strategy C. Under weak demand, revenue is 210 for Strategy A, 230 for Strategy B, and 220 for Strategy C. Operating costs, in thousands of dollars, depend on the fuel cost level and strategy. If fuel costs are low, operating costs are 160 for Strategy A, 190 for Strategy B, and 210 for Strategy C. If fuel costs are high, operating costs are 220 for Strategy A, 250 for Strategy B, and 290 for Strategy C. The cooperative should treat each combined demand and fuel-cost outcome as a state of nature and use the stated probabilities to compute expected profits.

Problem 3: A community sports center must choose a membership model for the next year: Model A (standard monthly pass), Model B (premium pass with classes), or Model C (flex pass). Demand for memberships can be high, medium, or low, and staffing costs can be favorable or unfavorable. Demand probabilities are $0{,}25$ (high), $0{,}50$ (medium), and $0{,}25$ (low). Staffing costs are independent of demand, with probabilities $0{,}60$ (favorable) and $0{,}40$ (unfavorable). Expected membership counts depend on demand and model: under high demand the center expects 620 members with Model A, 480 with Model B, and 700 with

Model C; under medium demand it expects 460 with Model A, 360 with Model B, and 520 with Model C; under low demand it expects 300 with Model A, 240 with Model B, and 360 with Model C. Membership fees are 35 for Model A, 48 for Model B, and 30 for Model C per member per year. Variable staffing cost per member is 18 when costs are favorable and 26 when costs are unfavorable. Fixed annual costs are $6{,}500$ for Model A, $9{,}000$ for Model B, and $5{,}200$ for Model C. All monetary amounts in this point can be treated in dollars. Use the combined demand and staffing outcomes as states of nature and the stated probabilities to compute expected profits.